

Benchmark #1

2MeV Fe on Fe

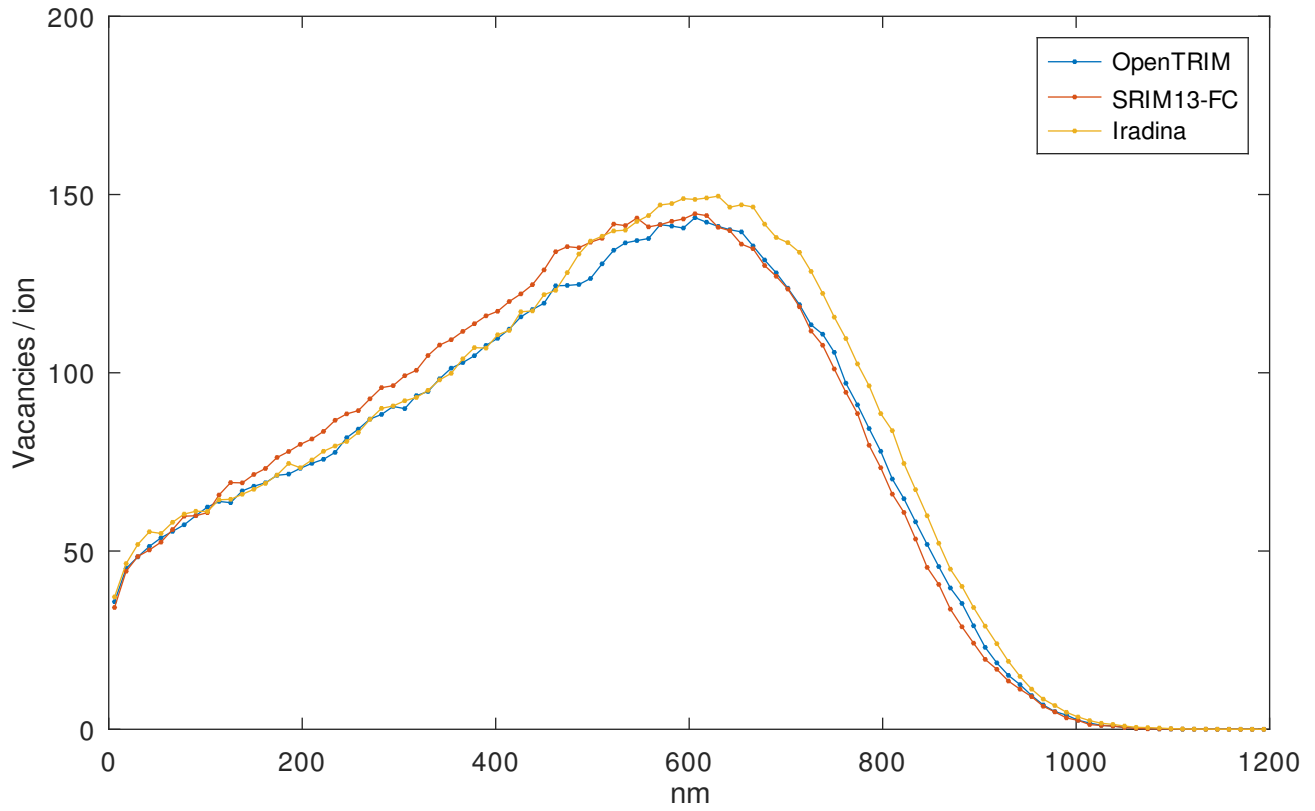
Ion energy $E_0 = 2e+06$ eV

Target depth = 1200 nm

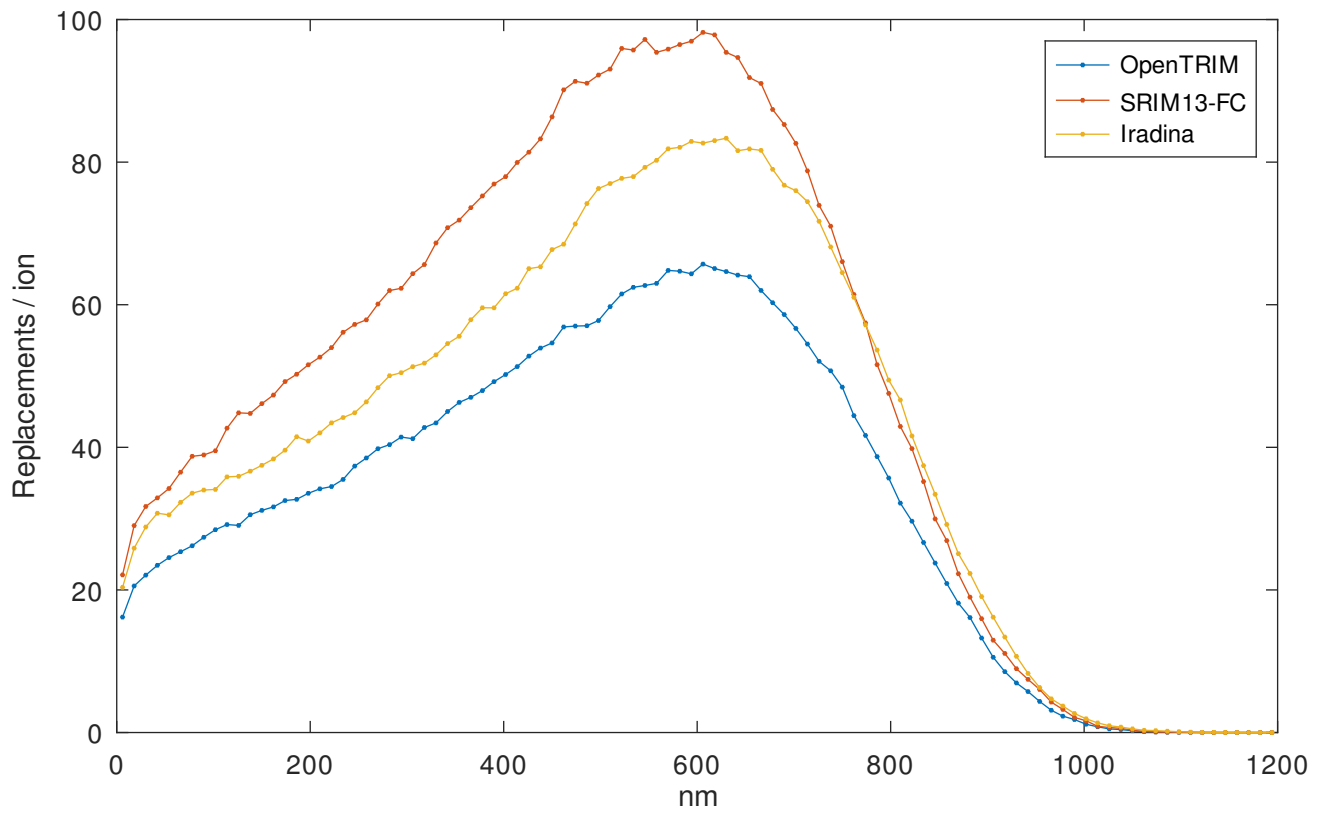
Summary Table

Quantity	OpenTRIM	SRIM13-FC	Iradina
$V(Fe)$	7.17e+03	7.35e+03	7.53e+03
$R(tot)$	3.28e+03	4.87e+03	4.19e+03
$I(Fe)$	1	1	1
$\Delta Ee(Fe)/E_0$	0.556	0.552	0
$\Delta Ee(r)/E_0$	0.105	0.112	0
$\Delta Ee(tot)/E_0$	0.662	0.663	0.681
$\Delta En(Fe)/E_0$	0.00348	0.0034	0
$\Delta En(r)/E_0$	0.335	0.325	0
$\Delta En(tot)/E_0$	0.338	0.329	0.319
$1 - (\Delta Ee + \Delta En)/E_0$	0.000268	0.00772	0.000285

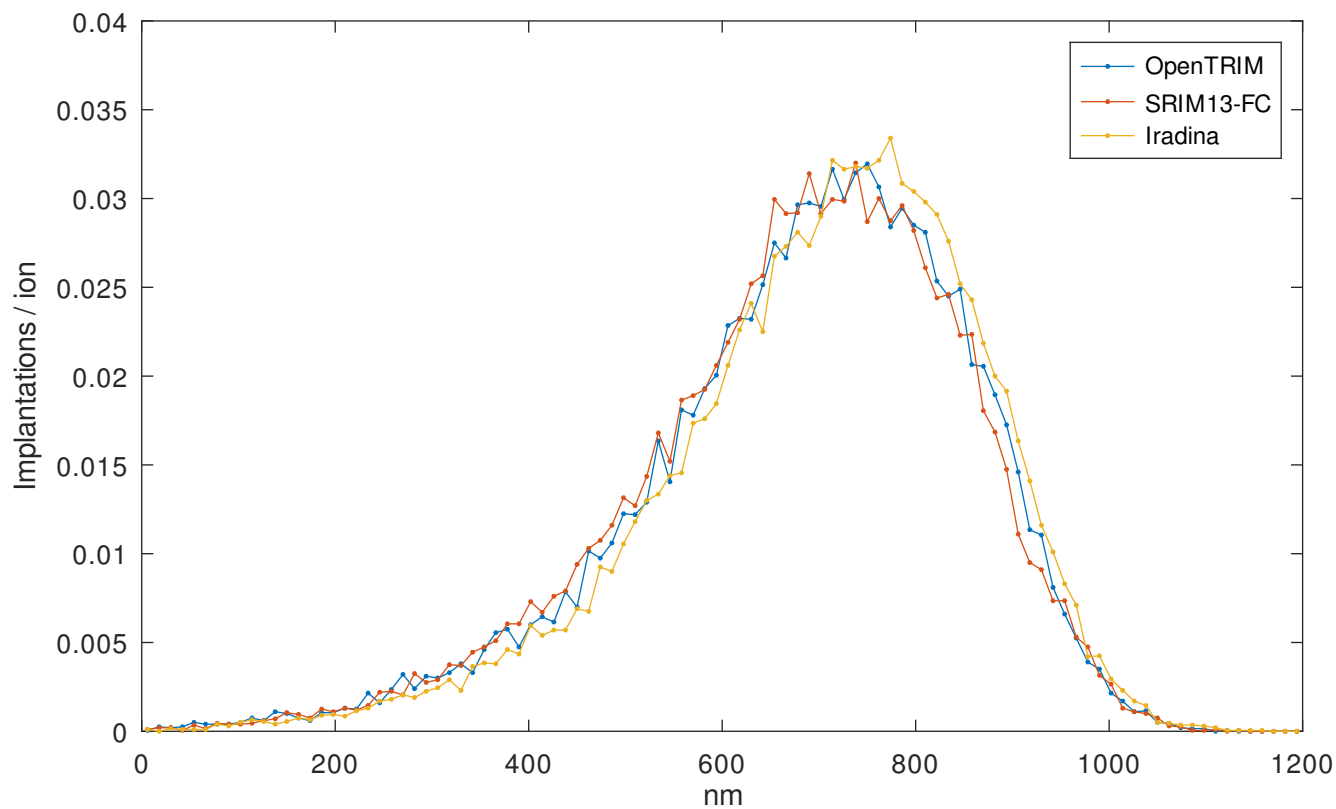
Vacancies of Fe in Fe



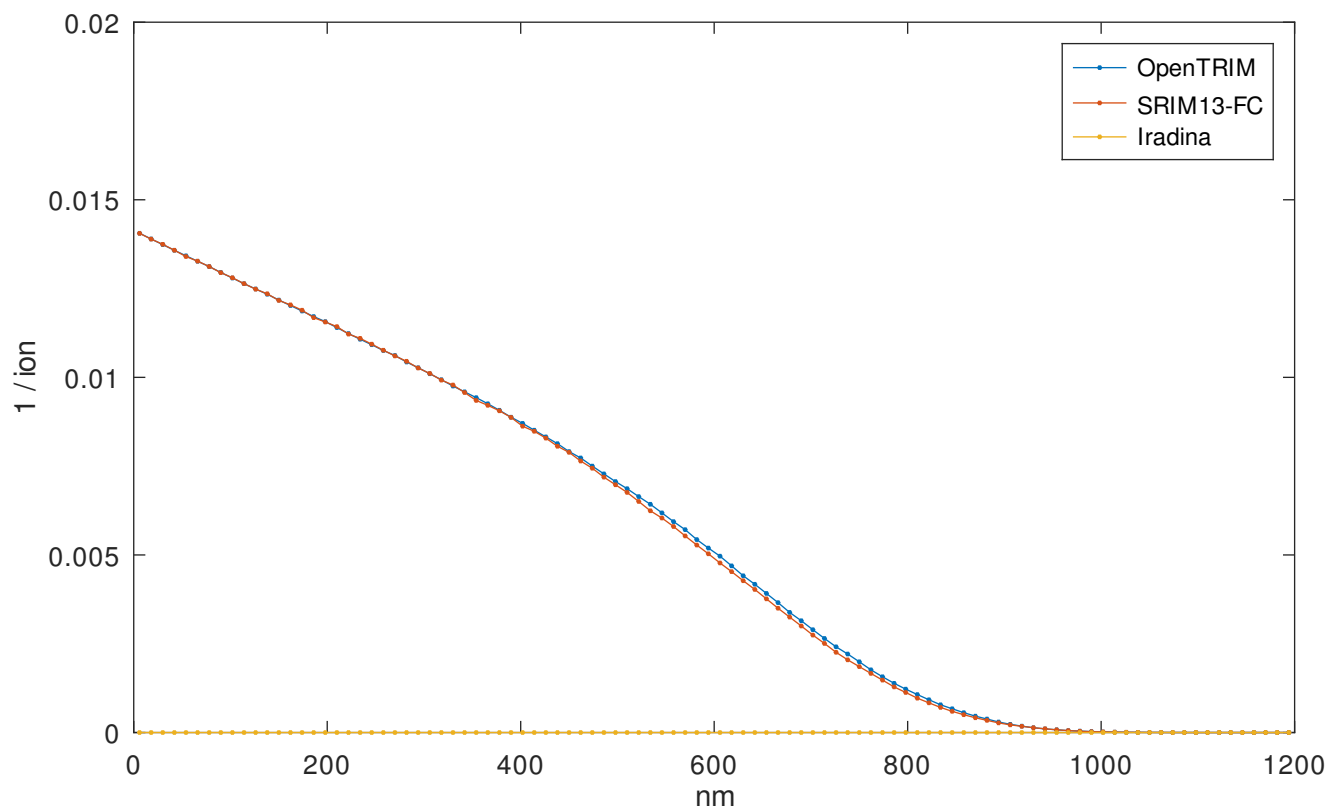
Replacements



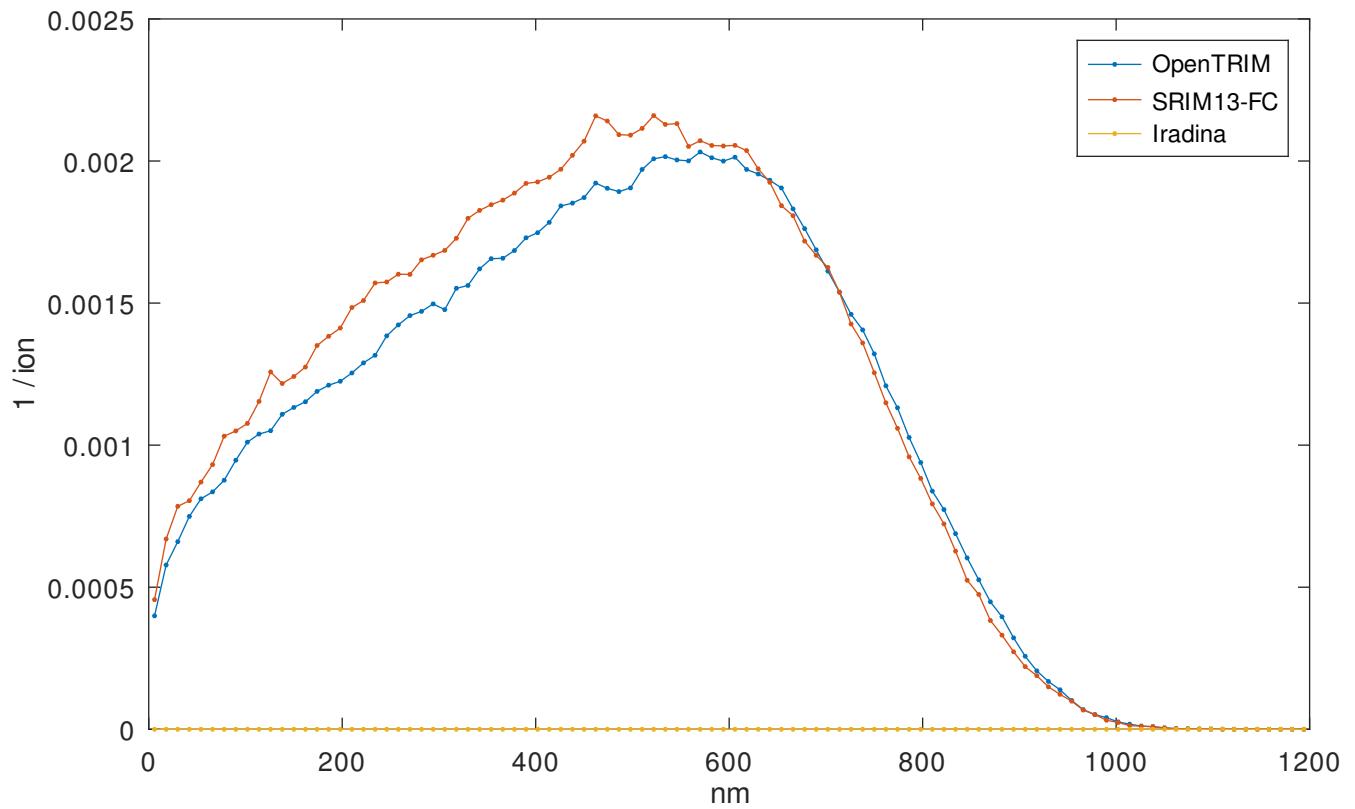
Implanted Fe ion



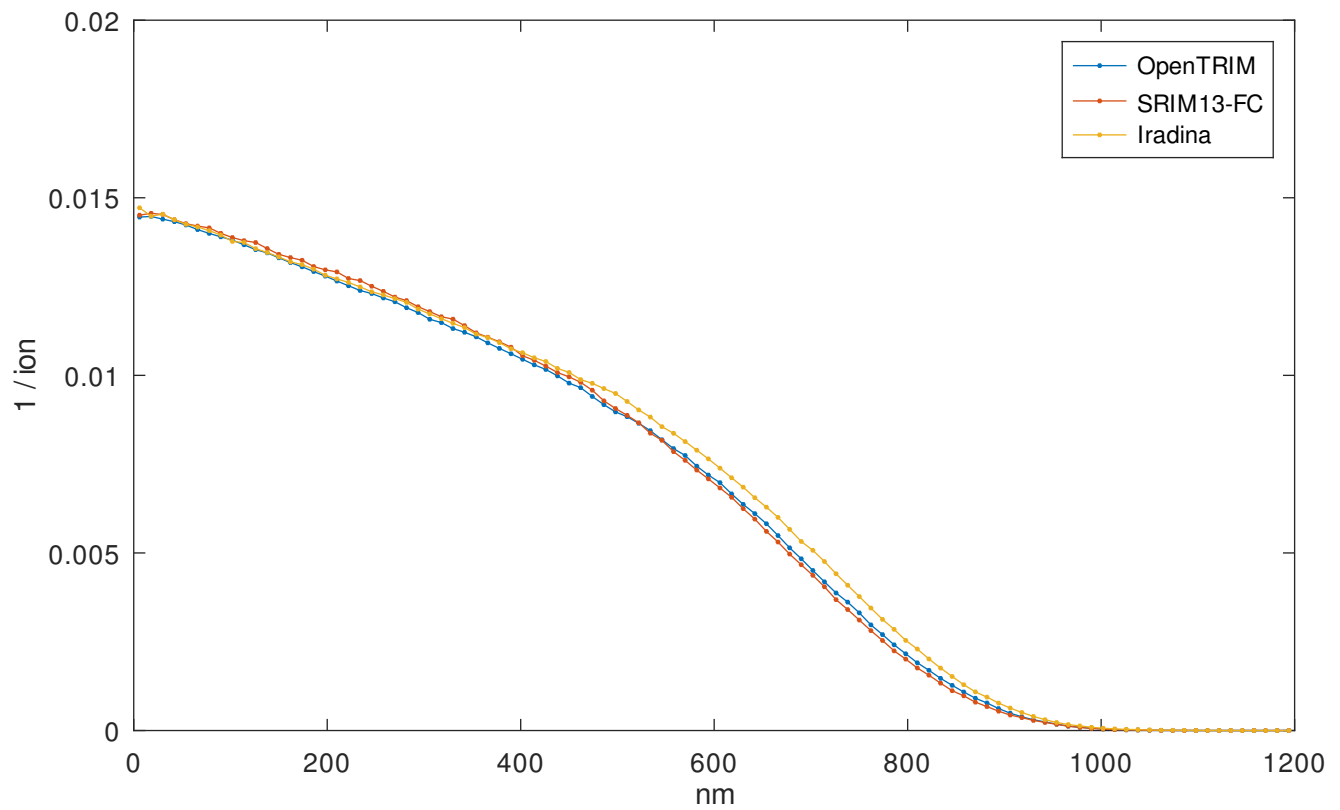
Electronic energy loss fraction $\Delta E_e/E_0$ by Fe ion



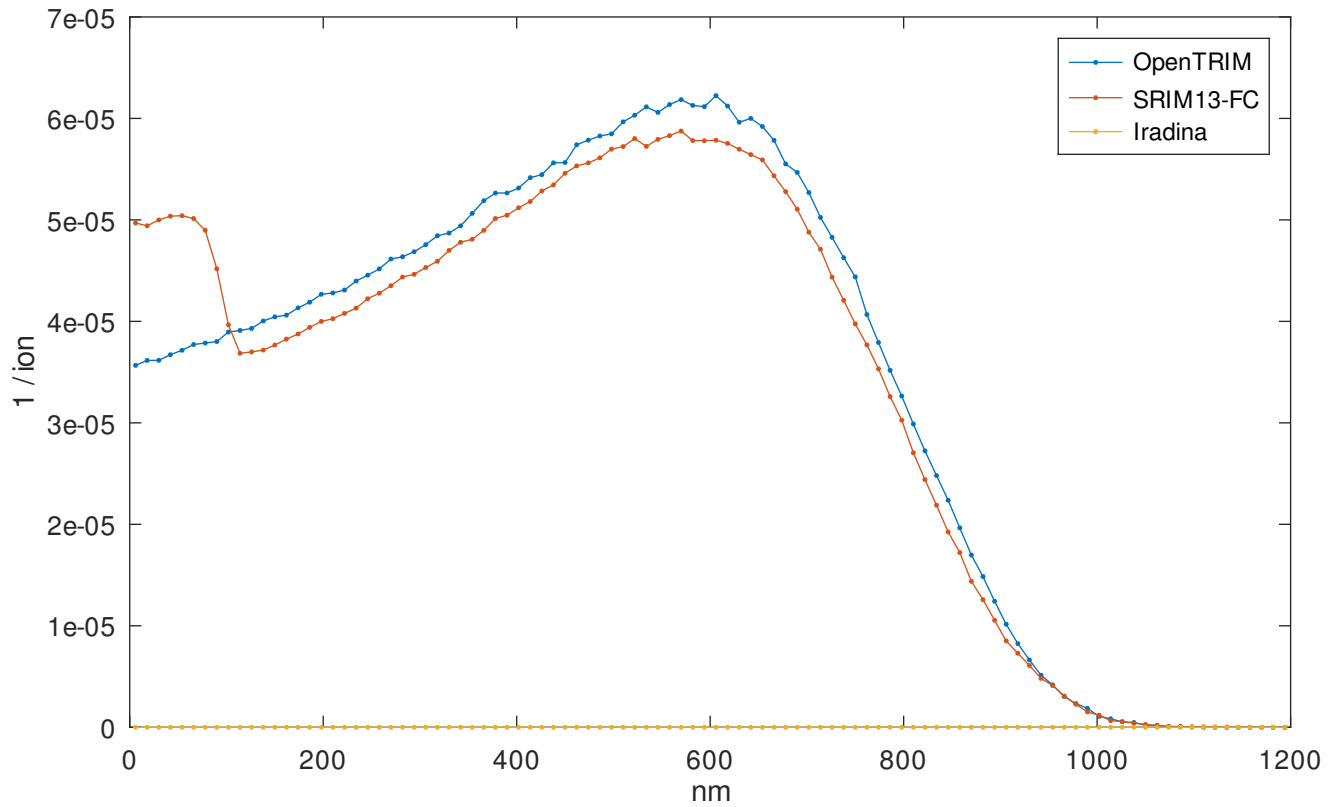
Electronic energy loss fraction $\Delta E_e/E_0$ by recoils



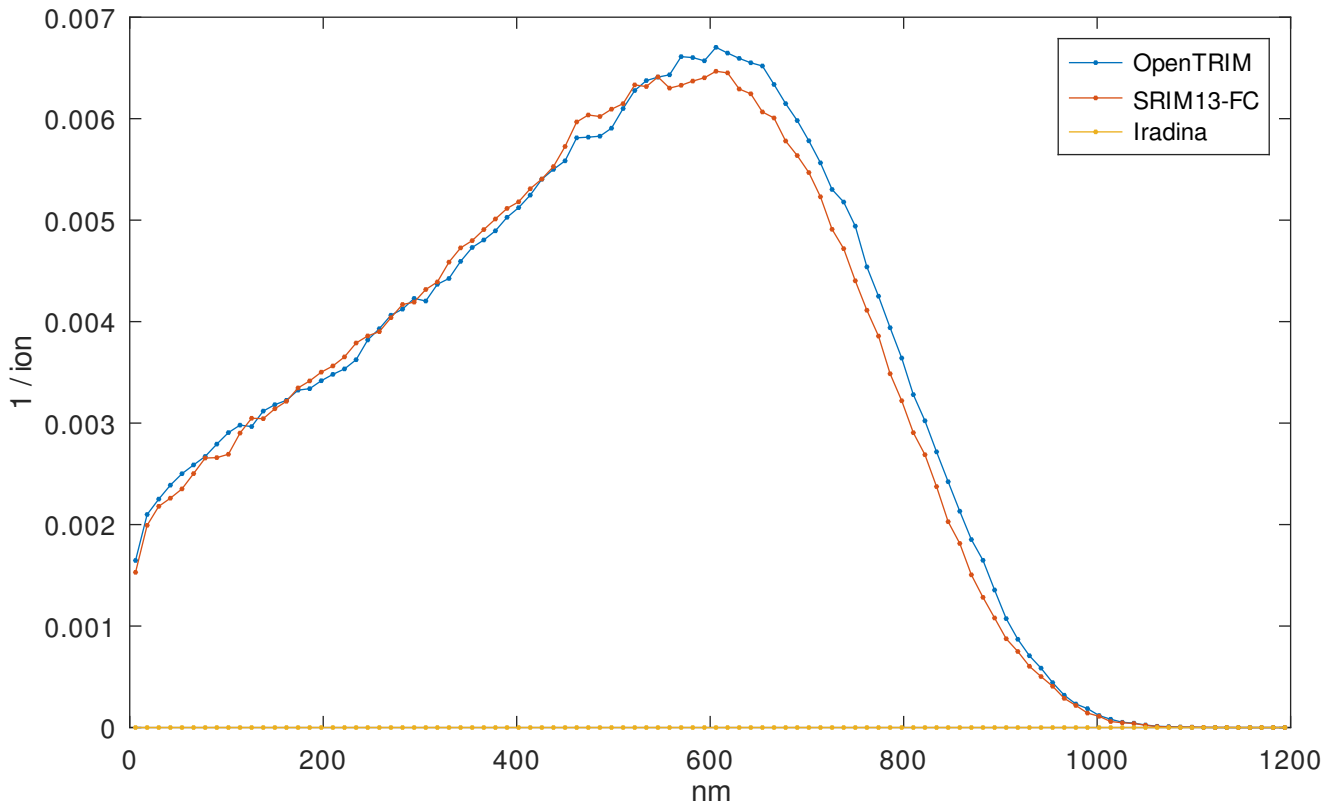
Total Electronic energy loss fraction $\Delta E_e/E_0$



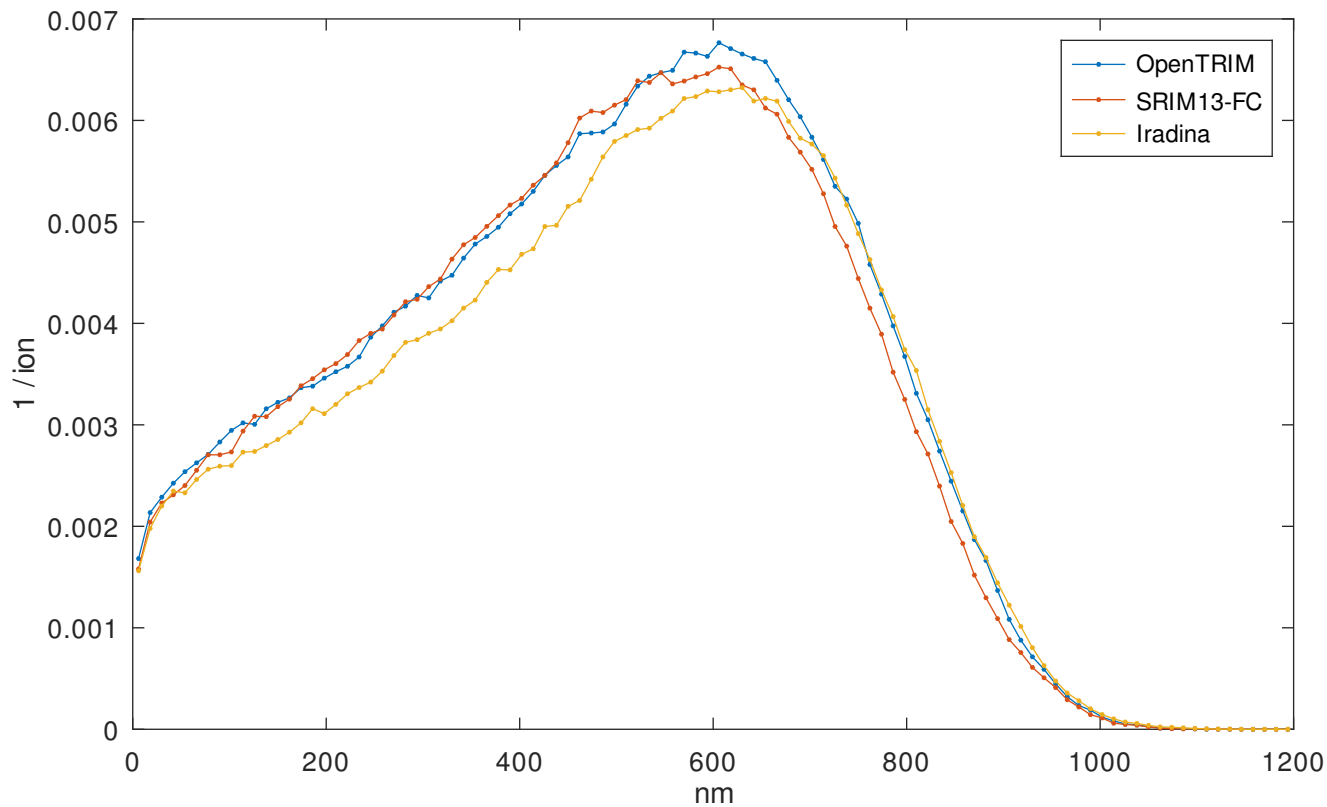
Nuclear energy loss fraction $\Delta E_n/E_0$ by Fe ion



Nuclear energy loss fraction $\Delta E_n/E_0$ by recoils



Total Nuclear energy loss fraction $\Delta E_n/E_0$



Total fractional energy loss $(\Delta E_e + \Delta E_n)/E_0$

